

***K*-means Clustering**

- Partitioning Clustering Approach
 - a typical clustering analysis approach via **iteratively** partitioning training data set to learn a partition of the given data space
 - learning a partition on a data set to produce several non-empty clusters (usually, the number of clusters given in advance)
 - in principle, optimal partition achieved via **minimising the sum of squared distance to its "representative object" in each cluster**

$$E = \sum_{k=1}^K \sum_{\mathbf{x} \in C_k} d^2(\mathbf{x}, \mathbf{m}_k)$$

e.g., Euclidean distance $d^2(\mathbf{x}, \mathbf{m}_k) = \sum_{n=1}^N (x_n - m_{kn})^2$

- Given a K , find a partition of K *clusters* to optimise the chosen partitioning criterion (cost function)
 - global optimum: exhaustively search all partitions
- The *K-means* algorithm: a heuristic method
 - K-means algorithm (MacQueen'67): each cluster is represented by the centre of the cluster and the algorithm converges to stable centriods of clusters.
 - K-means algorithm is the simplest partitioning method for clustering analysis and widely used in data mining applications.

K-means Algorithm

- Given the cluster number K , the *K-means* algorithm is carried out in three steps after initialisation:

Initialisation: set seed points (randomly)

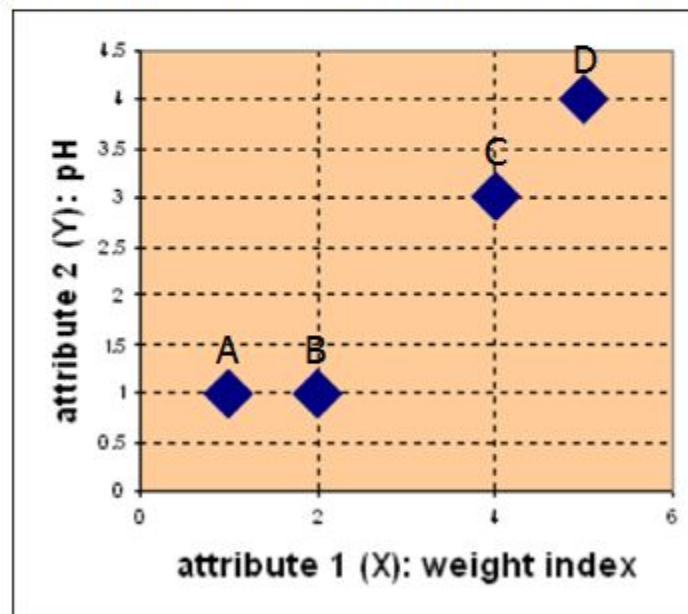
- 1) Assign each object to the cluster of the nearest seed point measured with a specific distance metric
- 2) Compute new seed points as the centroids of the clusters of the current partition (the centroid is the centre, i.e., *mean point*, of the cluster)
- 3) Go back to Step 1), stop when no more new assignment (i.e., membership in each cluster no longer changes)

Example

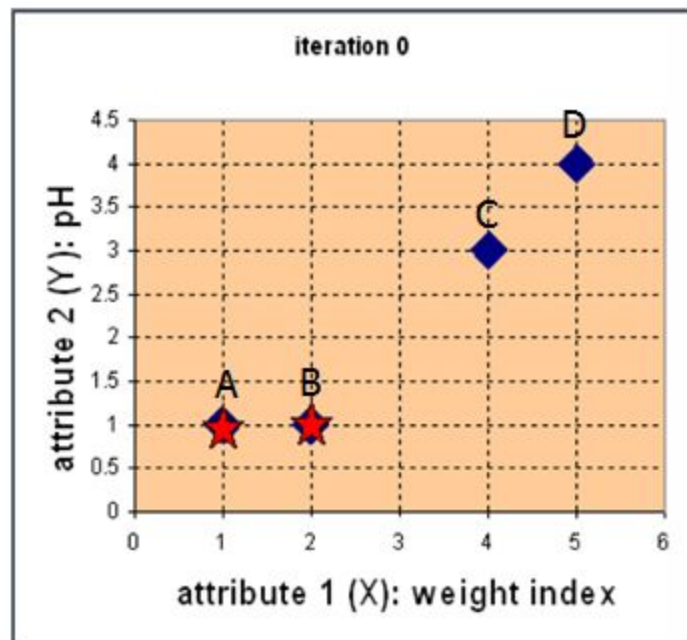
- Problem

Suppose we have 4 types of medicines and each has two attributes (pH and weight index). Our goal is to group these objects into $K=2$ group of medicine.

Medicine	Weight	pH-Index
A	1	1
B	2	1
C	4	3
D	5	4



Example



Step 1: Use initial seed points for partitioning

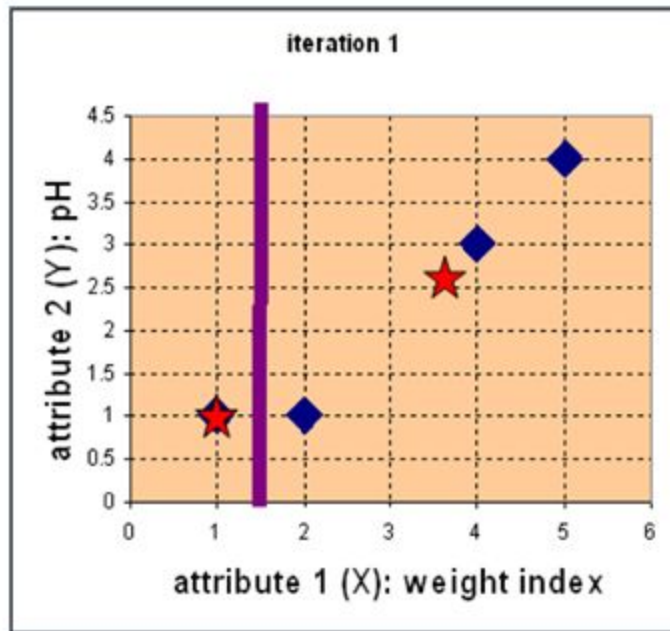
$$c_1 = A, c_2 = B$$

Step 2 Assign each object to the cluster with the nearest seed point

$$d(D, c_1) = \sqrt{(5-1)^2 + (4-1)^2} = 5$$

$$d(D, c_2) = \sqrt{(5-2)^2 + (4-1)^2} = 4.24$$

Step 3 Compute new centroids of the current partition



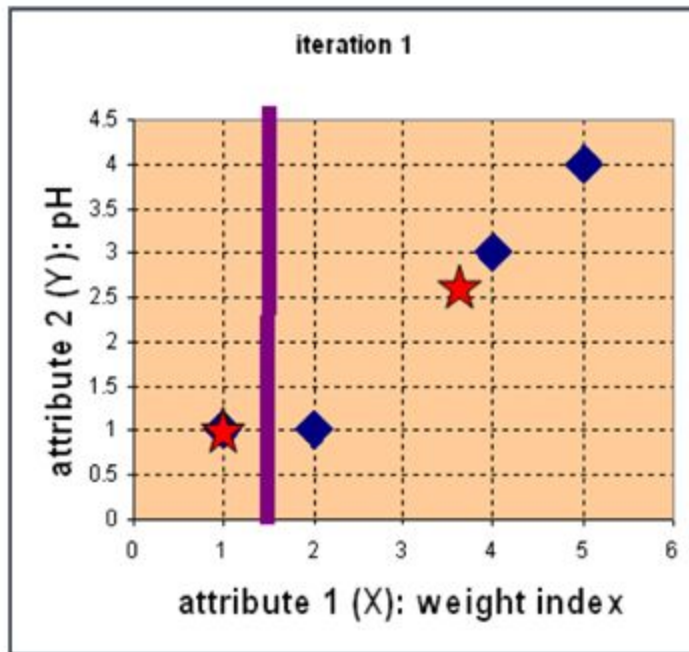
Knowing the members of each cluster, now we compute the new centroid of each group based on these new memberships.

$$c_1 = (1, 1)$$

$$c_2 = \left(\frac{2 + 4 + 5}{3}, \frac{1 + 3 + 4}{3} \right) \\ = \left(\frac{11}{3}, \frac{8}{3} \right)$$

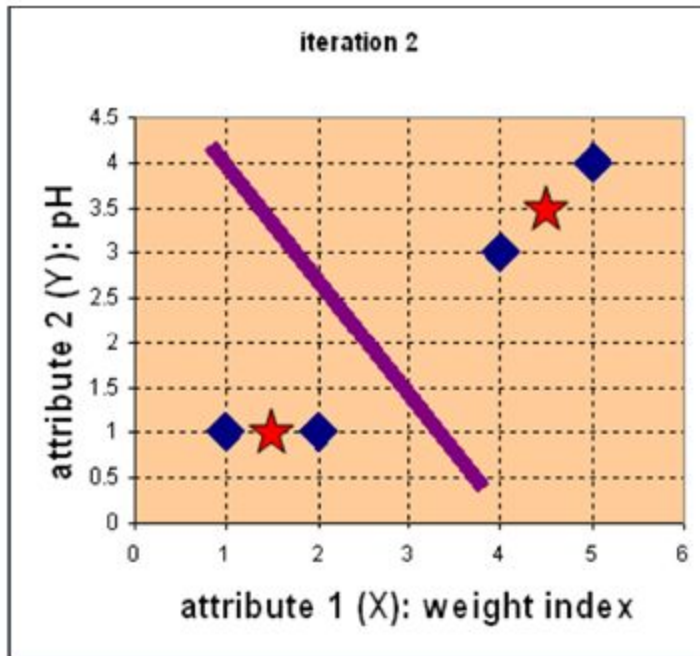
Step 2 Assign each object to the cluster with the nearest seed point

Renew membership based on new centroids



Step 3 Compute new centroids of the current partition

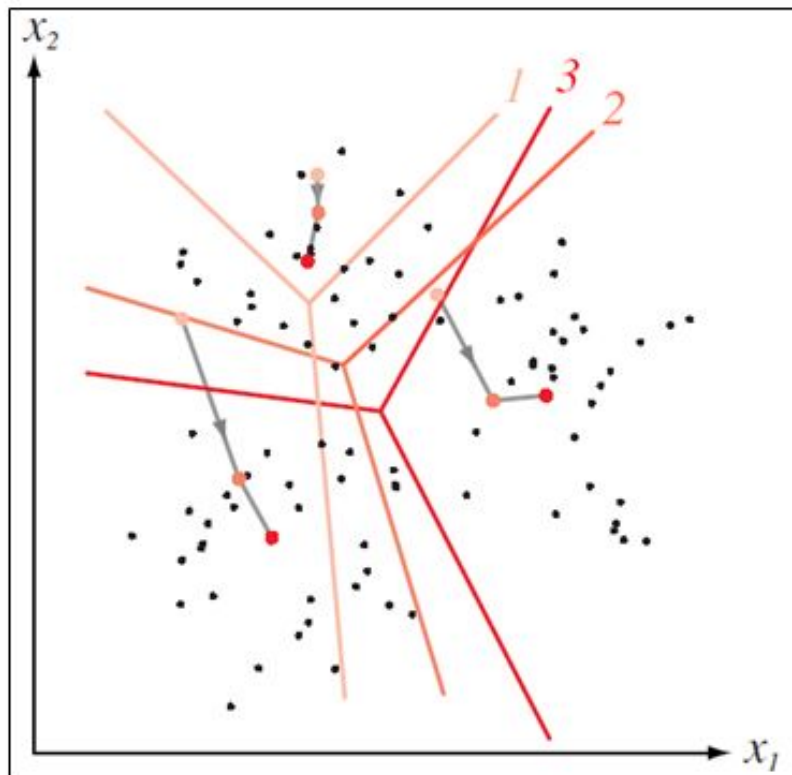
Repeat until its convergence



$$c_1 = \left(\frac{1+2}{2}, \frac{1+1}{2} \right) = \left(1\frac{1}{2}, 1 \right)$$

$$c_2 = \left(\frac{4+5}{2}, \frac{3+4}{2} \right) = \left(4\frac{1}{2}, 3\frac{1}{2} \right)$$

How K-means partitions?



When K centroids are set/fixed, they partition the whole data space into K mutually exclusive subspaces to form a partition.

A partition amounts to a

Voronoi Diagram

Changing positions of centroids leads to a new partitioning.

Relevant Issues

- Efficient in computation
 - $O(tKn)$, where n is number of objects, K is number of clusters, and t is number of iterations. Normally, $K, t \ll n$.
- Local optimum
 - sensitive to initial seed points
 - converge to a local optimum: maybe an unwanted solution
- Other problems
 - Need to specify K , the *number* of clusters, in advance
 - Unable to handle noisy data and outliers (*K-Medoids* algorithm)
 - Not suitable for discovering clusters with non-convex shapes
 - Applicable only when mean is defined, then what about categorical data? (*K-mode* algorithm)
 - how to evaluate the *K*-mean performance?

Summary

- **K-means** algorithm is a simple yet popular method for clustering analysis
- Its performance is determined by initialisation and appropriate distance measure
- There are several **variants** of *K*-means to overcome its weaknesses
 - K-Medoids: resistance to noise and/or outliers
 - *K*-Modes: extension to categorical data clustering analysis
 - CLARA: extension to deal with large data sets
 - Mixture models (EM algorithm): handling uncertainty of clusters